

SUBHAM DANGAR

M.Sc. Data Science Student

+91 7029325530 — dangarsubham@gmail.com — [LinkedIn](#) — [GitHub](#) — Bankura, West Bengal

PROFESSIONAL SUMMARY

M.Sc. Data Science student at RKMVERI with a strong mathematical foundation and hands-on experience in *Machine Learning, Deep Reinforcement Learning, and distributed AI systems*. B.Sc. Mathematics (Hons) graduate with CGPA 8.01. Built end-to-end ML pipelines, deployed APIs, and developed AI-powered semantic search systems independently.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, R, SQL

ML / DL: Supervised Learning, Deep Q-Networks, Semantic Search, Model Evaluation

Libraries & Frameworks: NumPy, Pandas, Matplotlib, scikit-learn, sentence-transformers, Dask, FastAPI

Databases & Tools: SQLite, Git, GitHub, Jupyter, Google Colab, VS Code, Excel

PROJECTS

Distributed Educational Video Search System

Ongoing

[GitHub Repo](#)

- Developed an AI-powered system to retrieve top 3 relevant YouTube timestamps from multilingual user queries.
- Used `sentence-transformers` and cosine similarity for semantic retrieval; ranked results by relevance and view count.
- Implemented parallel search with Dask, a semantic cache (SQLite), and a web fallback mechanism for reliability.

Spam Email Detection System

Sep 2025 – Nov 2025

Python, scikit-learn, TF-IDF, Flask [GitHub Repo](#)

- Built end-to-end ML pipeline with text preprocessing and TF-IDF feature extraction.
- Trained and compared Naive Bayes, Logistic Regression, and Random Forest models.
- Evaluated using precision, recall, F1-score, and ROC-AUC; deployed model via Flask for real-time prediction.

Mountain Car RL Agent (DQN)

Sep 2025 – Nov 2025

Python, Gymnasium, Deep Q-Network [GitHub Repo](#)

- Implemented Deep Q-Network with experience replay, target network, and epsilon-greedy exploration strategy.
- Tuned hyperparameters and tracked learning curves to achieve stable convergence in the Gymnasium environment.

ACADEMIC RESEARCH

A Study on Countability — B.Sc. Dissertation

Feb 2025 – Jun 2025

[GitHub Repo](#)

- Studied finite, countable, and uncountable sets with formal proofs: Cantor's diagonal argument, countability of $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$.

EDUCATION

M.Sc. Data Science — Jun 2025 – Present — RKMVERI, Department of Computer Science

B.Sc. Mathematics (Hons) — Jun 2022 – May 2025 — Bankura Christian College **CGPA: 8.01 / 10**

Higher Secondary (XII) — 2022 — WBCHSE **94.8%**

Secondary (X) — 2020 — WBBSE **92.14%**

CERTIFICATIONS & SCHOLARSHIP

Python Programming – NPTEL — [View Certificate](#) **C++ Programming – NPTEL** — [View Certificate](#)

INSPIRE Scholarship – Govt. of India (Merit-based) — [View Offer Letter](#)